

# The empirical recurrence for OEIS A301658, proved by transfer matrix

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## Abstract

OEIS A301658 counts 4-column arrays over  $\{0, \dots, 1\}$  in which every entry has, among the four horizontal and vertical neighbours, a number *equalto* itself lying in  $\{0, 1, 4\}$ , and it carries an empirical recurrence of order 5 contributed by R. H. Hardin. It is true. The neighbourhood reaches one row up and one row down, so the condition on a row involves three consecutive rows and a window of two rows is a sufficient state; the array is then a walk in a digraph on  $S = 256$  vertices. Unlike a plain walk count the boundary rows matter here — the first row has nothing above it and the last nothing below — so the count is  $\mathbf{s}^\top P^{n-2} \mathbf{e}$  for genuine start and end vectors rather than all-ones. The conjectured recurrence is then annihilated exactly, in integer arithmetic.

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## 1 The sequence and the conjecture

OEIS A301658 is “Number of  $n \times 4$  0..1 arrays with every element equal to 0, 1 or 4 horizontally or vertically adjacent elements, with upper left element zero.”. It has offset 1 and begins

$$5, 6, 9, 13, 24, 45, 89, 182, \dots$$

The entry carries the following comment:

$$\text{Empirical: } a(n) = 2*a(n-1) + a(n-2) - a(n-3) - 2*a(n-4) + a(n-5)$$

As of the “Last modified” line on the live entry (Mar 25 2018 11:26:05, revision 4) the statement is still recorded as empirical, and nothing on the entry records it as settled either way.

Written out, the claim is

$$a(n) = 2a(n-1) + a(n-2) - a(n-3) - 2a(n-4) + a(n-5) \tag{1}$$

for all  $n > 5$ .

## 2 The array as a walk, with boundary

For a cell of the array let its *score* be the number of the four horizontal and vertical neighbours that lie inside the array and carry a value *equalto* the cell’s own. The entry’s condition is that every cell has score in  $\{0, 1, 4\}$ . The entry also fixes the top-left entry to be 0; that is a condition on the first row alone, so it restricts the start vector and nothing else.

A cell’s neighbourhood reaches at most one row up and one row down, so the scores of the cells of a row  $s$  are determined by  $s$  together with the row above and the row below. Take as state an ordered pair  $(r, s)$  of rows, so  $S = 256$ , and set

- $\mathbf{s}(r, s) = 1$  when every cell of  $r$  has score in  $\{0, 1, 4\}$  with no row above  $r$  and  $r_1 = 0$ , and 0 otherwise;
- an edge  $(r, s) \rightarrow (s, t)$  when every cell of  $s$  has score in  $\{0, 1, 4\}$  computed from  $r, s$  and  $t$ ;
- $\mathbf{e}(r, s) = 1$  when every cell of  $s$  has score in  $\{0, 1, 4\}$  with no row below  $s$ , and 0 otherwise.

Let  $P$  be the adjacency matrix of that edge relation.

**Lemma 1.**  $a(n) = \mathbf{s}^\top P^{n-2} \mathbf{e}$  for every  $n \geq 2$ .

*Proof.* An array with  $n$  rows is a sequence  $r^{(1)}, \dots, r^{(n)}$  of rows, and it is counted exactly when every one of its  $n$  rows has all scores in  $\{0, 1, 4\}$ . Read the sequence as the walk  $(r^{(1)}, r^{(2)}) \rightarrow (r^{(2)}, r^{(3)}) \rightarrow \dots \rightarrow (r^{(n-1)}, r^{(n)})$ , of length  $n - 2$ . The condition on the first row is checked by  $\mathbf{s}$  at the initial state, because a first row has no row above it; the condition on row  $i$  for  $2 \leq i \leq n - 1$  is checked by the edge entering  $(r^{(i)}, r^{(i+1)})$ , which sees exactly  $r^{(i-1)}, r^{(i)}$  and  $r^{(i+1)}$ ; and the condition on the last row is checked by  $\mathbf{e}$  at the final state. Each row's condition is therefore tested once and with the correct neighbours, and summing over all walks gives  $\mathbf{s}^\top P^{n-2} \mathbf{e}$ .  $\square$

### 3 The criterion

Let  $q(t) = t^5 - \sum_i c_i t^{5-i}$  be the characteristic polynomial of (1) and  $u_j = \mathbf{s}^\top P^j q(P) \mathbf{e}$ .

**Lemma 2.**  $a(n) - \sum_i c_i a(n - i) = u_{n-2-5}$  whenever  $n - 5 \geq 2$ . So (1) holds for every  $n > t$  exactly when  $u_j = 0$  for all  $j > t - 2 - 5$ . Moreover  $(u_j)$  satisfies the monic linear recurrence given by the characteristic polynomial of  $P$ , of degree  $S$ , so  $S$  consecutive zeros force every later one and the last nonzero  $u_j$  pins  $t$  down exactly.

*Proof.* Substituting Lemma 1 and factoring  $P^{n-2-5}$  out of  $P^{n-2} - \sum_i c_i P^{n-i-2}$  gives the identity, and letting the exponent run gives the equivalence. For the last part, Cayley–Hamilton gives  $\chi(P) = 0$  for the characteristic polynomial  $\chi$ , so  $\chi$  applied to the shift annihilates  $(u_j)$ ;  $\chi$  is monic of degree  $S$ , which propagates  $S$  consecutive zeros forward.  $\square$

### 4 The computation

**Theorem 3.** The recurrence (1) holds for every  $n > 5$ . This is the statement of Section 1.

*Proof.* The vector  $q(P) \mathbf{e} \in \mathbb{Z}^S$  was formed by 5 matrix–vector products in exact integer arithmetic and the  $u_j$  evaluated against  $\mathbf{s}$  in turn. They vanish from the index corresponding to  $n = 5 + 1$  onwards, and the run was continued past  $S$  consecutive zeros, which by Lemma 2 settles every larger index.  $\square$

### 5 Verification

Three checks, each independent of the algebra above.

First, the digraph, the start vector and the end vector were built directly from the entry's stated condition, the one-row count computed separately, and the resulting values compared against the entry's DATA at every one of the 32 published indices; they agree exactly. This is the check that ties the model to the sequence, and it is a sharp one: getting the boundary rows wrong, or treating them as interior rows, changes the early terms immediately.

Second, the recurrence (1) was evaluated on those published terms in exact integer arithmetic, with no matrices involved, and vanishes at every index the entry claims.

Third, the vanishing test of Lemma 2 was carried past  $S = 256$  consecutive zeros rather than to a sampled prefix.

## References

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